

1.4 Network security

How to run this unit, lesson by lesson

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Start here

12 lessons. A realistic estimate is **12 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Threat map ns-101	1	Lesson_Plans/NS_L01_LessonPlan.pdf
2	Malware lab ns-102	1	Lesson_Plans/NS_L02_LessonPlan.pdf
3	Database break-in ns-103	1	Lesson_Plans/NS_L03_LessonPlan.pdf
4	Phishing inbox ns-104	1	Lesson_Plans/NS_L04_LessonPlan.pdf
5	Social engineering HQ ns-105	1	Lesson_Plans/NS_L05_LessonPlan.pdf
6	Cracking lab ns-106	1	Lesson_Plans/NS_L06_LessonPlan.pdf
7	Traffic flood ns-107	1	Lesson_Plans/NS_L07_LessonPlan.pdf
8	Packet sniffers ns-108	1	Lesson_Plans/NS_L08_LessonPlan.pdf
9	Defence depot ns-109	1	Lesson_Plans/NS_L09_LessonPlan.pdf
10	Defence depot 2 ns-110	1	Lesson_Plans/NS_L10_LessonPlan.pdf
11	End-of-topic test ns-111-test	1	Lesson_Plans/NS_L11_TEST_LessonPlan.pdf
12	Cyber defence ns-bonus	1	Lesson_Plans/NS_BONUS_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (`teacher.html`), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at `topics.html` with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- only one activity asks the pupil to produce an answer rather than choose one — ns-101, ns-102, ns-103, ns-104, ns-105, ns-106, ns-107, ns-108, ns-110
- no optional challenge activity — ns-101, ns-102, ns-103, ns-104, ns-105, ns-106, ns-107, ns-108, ns-109, ns-110
- every activity is recognition: the answer is always on the screen — ns-109

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Threat map ns-101

What forms of attack threaten a network, and what do attackers want?

OUTCOMES	understand the different forms of attack against computer systems
STATIONS AND ACTIVITIES	6 stations, 11 activities, 23 marks (10 guided, 1 independent, 0 challenge)
RESOURCES	NS_L01_ThreatMap_Lesson.pptx, NS_L01_ThreatMap_Worksheet.docx
PLAN	Lesson_Plans/NS_L01_LessonPlan.pdf

2. Malware lab ns-102

How do the different types of malware spread, and what damage do they do?

OUTCOMES	understand the threats of various forms of malware; understand how to identify and protect against malware
STATIONS AND ACTIVITIES	6 stations, 12 activities, 19 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	NS_L02_MalwareLab_Lesson.pptx, NS_L02_MalwareLab_Worksheet.docx
PLAN	Lesson_Plans/NS_L02_LessonPlan.pdf

3. Database break-in ns-103

How does an SQL injection attack work, and how can it be prevented?

OUTCOMES	understand the concept of SQL injection; understand how to protect against SQL injection
STATIONS AND ACTIVITIES	6 stations, 12 activities, 19 marks (11 guided, 1 independent, 0 challenge)

RESOURCES	NS_L03_DatabaseBreakIn_Lesson.pptx, NS_L03_DatabaseBreakIn_Worksheet.docx
PLAN	Lesson_Plans/NS_L03_LessonPlan.pdf

4. Phishing inbox ns-104

How can you recognise a phishing message, and what should you do about it?

OUTCOMES	understand what phishing is; understand how to identify and protect against phishing
STATIONS AND ACTIVITIES	6 stations, 12 activities, 19 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	NS_L04_PhishingInbox_Lesson.pptx, NS_L04_PhishingInbox_Worksheet.docx
PLAN	Lesson_Plans/NS_L04_LessonPlan.pdf

5. Social engineering HQ ns-105

What does a phishing attack set out to achieve?

OUTCOMES	understand what phishing is; understand how to identify and protect against phishing; understand the ways people are a weak point
STATIONS AND ACTIVITIES	6 stations, 12 activities, 22 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	NS_L05_SocialEngineeringHQ_Lesson.pptx, NS_L05_SocialEngineeringHQ_Worksheet.docx
PLAN	Lesson_Plans/NS_L05_LessonPlan.pdf

6. Cracking lab ns-106

How does a brute-force attack work?

OUTCOMES	understand what a brute-force attack is; understand how to identify and protect against brute-force attacks
STATIONS AND ACTIVITIES	6 stations, 13 activities, 18 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NS_L06_CrackingLab_Lesson.pptx, NS_L06_CrackingLab_Worksheet.docx
PLAN	Lesson_Plans/NS_L06_LessonPlan.pdf

7. Traffic flood ns-107

How does a denial-of-service attack work, and how can it be defended against?

OUTCOMES	understand what a denial-of-service attack is; understand how to identify and protect against denial-of-service attacks
STATIONS AND ACTIVITIES	6 stations, 13 activities, 17 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NS_L07_TrafficFlood_Lesson.pptx, NS_L07_TrafficFlood_Worksheet.docx
PLAN	Lesson_Plans/NS_L07_LessonPlan.pdf

8. Packet sniffers ns-108

How is data captured in transit, and how can it be protected?

OUTCOMES	understand the threat that data interception and theft poses; understand how to identify and protect against data interception
STATIONS AND ACTIVITIES	6 stations, 12 activities, 19 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	NS_L08_PacketSniffers_Lesson.pptx, NS_L08_PacketSniffers_Worksheet.docx
PLAN	Lesson_Plans/NS_L08_LessonPlan.pdf

9. Defence depot ns-109

Which protections stop which attacks, and why does no single one stop everything?

OUTCOMES	understand the ways people are a weak point in system security; know how penetration testing, anti-malware, firewalls and user access levels help against cyberattack
STATIONS AND ACTIVITIES	6 stations, 12 activities, 19 marks (12 guided, 0 independent, 0 challenge)
RESOURCES	NS_L09_DefenceDepot_Lesson.pptx, NS_L09_DefenceDepot_Worksheet.docx
PLAN	Lesson_Plans/NS_L09_LessonPlan.pdf

10. Defence depot 2 ns-110

How can we protect ourselves against hackers?

OUTCOMES	know how passwords, encryption and physical security help against cyberattack; be able to choose suitable protections for a given threat
STATIONS AND ACTIVITIES	6 stations, 12 activities, 24 marks (11 guided, 1 independent, 0 challenge)
RESOURCES	NS_L10_DefenceDepot2_Lesson.pptx, NS_L10_DefenceDepot2_Worksheet.docx
PLAN	Lesson_Plans/NS_L10_LessonPlan.pdf

11. End-of-topic test ns-111-test

Your 1.4 test is done on paper. Warm up with the Cyber defence game, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	NS_L11_EndOfTopicTest_Reflection_Worksheet.docx
PLAN	Lesson_Plans/NS_L11_TEST_LessonPlan.pdf

12. Cyber defence ns-bonus

Spend a budget on protections, identify each incoming threat and find out whether your defences hold. Three lives, then the network shuts down.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.
STATIONS AND ACTIVITIES	0 stations, 1 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	none recorded
PLAN	Lesson_Plans/NS_BONUS_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.